

Henry S. Trowbridge (喬念唐)

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Education

Bachelor of Science in Computer and Information Science; Bachelor of Arts in Chinese

The Ohio State University: Columbus, OH

Specialization: *Information & Computational Assurance*

August 2022 – December 2026

- GPA: 3.86 (Dean's List all completed terms)
- Student member of ACM (Member ID: 7641930, htrowbridge@acm.org)
- Relevant Coursework: Java, Operating Systems, Web Applications, Data Structures and Algorithms, Information Security

Work Experience

IT Intern – Enterprise Cybersecurity Policy

Chicago, IL

Abbott Laboratories

May 2026 – August 2026

- Worked on the technical evaluation of cloud-integrated AI transcription models for the ADC Lighthouse call-center Copilot application, engineering novel, production-compatible metrics to optimize data capture and customer triage
- Architected and delivered a data-backed business-case proposal to Global IT and executive leadership, defending strategic recommendations of optimal AI transcription models to elevate customer experience
- Led the end-to-end proof-of-concept for an internal cyber policy AI assistant; launched a cross-functional beta-test user study to optimize agent accuracy based on real-world stakeholder feedback
- Authored robust, reproducible engineering documentation for Copilot agent development and lifecycle maintenance; built a structured evaluation test-set and question framework to audit and validate LLM output accuracy
- Strengthened corporate cybersecurity posture by overhauling global enterprise cryptographic controls, aligning with modern NIST benchmarks and internal formatting expectations
- Partnered with cross-functional business stakeholders and subject matter experts to execute a preliminary gap analysis mapping internal cybersecurity controls directly to authoritative NIST secure software development lifecycle (SSDL) standards
- Accelerated the global control-objective modernization effort and strategy by cleaning up legacy controls down to reduce redundancies, ease adoption, and maximize auditability
- Designed and proposed a global training strategy and avenues of its delivery to scale cyber policy compliance across international business units
- Designed and engineered an automated financial data pipeline to aggregate disjointed cost-center statements into a unified budget dashboard, delivering critical financial visibility to leadership and eliminating hours of manual monthly labor

Undergraduate Research Assistant (Industrial Cybersecurity Lab)

Columbus, OH

Center for Design and Manufacturing Excellence

May 2025 – Present

- Co-led development of an [interactive web app](#) to help small manufacturers achieve [CMMC 2.0 cybersecurity compliance](#) needed for DoD contracts
- Developed a full-stack web app using container technologies, a custom database with role-based access controls (RBAC), and Python APIs for backend functionality; managed with Agile/Scrum and frequent client meetings
- Ran blockchain (hyperledger fabric) pruning and performance-measuring experiments

Software and Data Engineer Intern

Washington, D.C. (Remote)

GovRat

September 2025 – Present

- Developed Python-based automation to ingest and structure federal contracting data
- Ensured data reliability and consistency for in-house LLM (Large Language Model) training, directly enhancing the platform's ability to draft compliance-ready government-contract proposals

Classroom and Digital Union Support Technician

Columbus, OH

The Ohio State University Office of Technology and Digital Innovation

March 2025 – September 2025

- Resolved mission-critical IT/AV failures in classrooms, ensuring uninterrupted instruction for hundreds of students weekly
- Delivered faculty training and troubleshooting support that boosted classroom tech uptime and reduced repeat service calls

Accounting Intern

Chicago, IL

Jeffrey W. Krol & Associates, Ltd.

June 2023 - August 2023; May 2024 - July 2024

- Migrated client data from QuickBooks Online to Desktop, creating standardized documentation that colleagues reused
- Built digital archival system and Excel dashboards, reducing reporting turnaround time and improving investment tracking

Technical Projects and Research

U.S. Cyber Command Research Competition: Cyber RECon 2026's Winning Paper

Fort Meade, MD

Semantic Shields: Automating Critical Infrastructure Defense via NLP-Driven Ransomware Profiling

April 2026

- [Led the development, writing, research, and submission](#) of a novel research paper (as first author), supported by [my lab](#) working with a cross-functional team
- Won all three categories of US Cyber Command's 2026 Cyber RECon research competition - overall best research, best research paper in the analyst category, and best presentation (poster and slide-deck)
- Argued for the adoption of NLP (Natural Language Processing) to protect US critical infrastructure

Debian Package Mirrors

June 2025 – February 2026

- Deployed Debian mirrors on FreeBSD (complete) and Debian with NGINX, provided reliable package access for users
- Automated synchronization with official repositories, ensuring updated packages with minimal downtime

Henry Codes - Personal Portfolio (Live-Hosted) | GitHub

January 2025 – Present

- Designed and self-hosted personal portfolio site (on Debian, NGINX, PHP) with Cloudflare -- secure, reproducible, Docker-containerized

Leadership and Extracurricular Activities (largely at Ohio State - Columbus, OH)

- **Vice President, [Cybersecurity Club](#)** | Nov 2022 – Present | Organized events (BuckeyeCTF, Ohio LinuxFest collab) and meetings
- **Vice President, [Open Source Club](#)** | Sep 2024 – Present | Advocated for Linux and open-source tools
- **Member, [History Club](#)** | Oct 2022 – Present | Learned history; presented on Chinese three-kingdoms periods and other topics
- **Consulting Member (Former President), [Buckeye Table Tennis Club](#)** | Aug 2022 – Present | Player and consultant for leadership

Skills

- **Professional:** Adaptability, Cross-Functional Collaboration, Client Communication, Project Management, Technical Documentation, Change Management, Agile/Scrum, English (Fluent), Chinese (Professional Working Proficiency)
- **Backend Development:** Python (FastAPI, Flask, SQLAlchemy, Alembic, Pandas, Plotly, REST APIs), Ruby (Ruby on Rails), Java, C#, C, SQL (PostgreSQL, SQLite)
- **Infrastructure & Security:** Governance, Risk, and Compliance; AI Strategy; AI Model Evaluation; Cyber Risk Management; Process Automation; CMMC (NIST 800-171r3), Linux (Debian, RHEL), FreeBSD, Docker, Podman, Git, NGINX, SSH, RDP, UFW, Firewalls, Webservers, Cloudflare, Tailscale, Zero Trust, Principle of Least Privilege, Access Control Lists, Role-Based Access Controls
- **Frontend Development:** HTML, CSS, JavaScript, Jinja, TailwindCSS